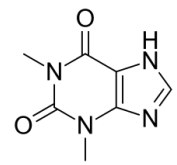
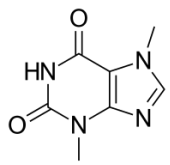
Analytical Chemistry
Cumulative Exam
April 4, 2025

Theme: Separations.



Theophylline

$C_7H_8N_4O_2$



Theobromine

$C_7H_8N_4O_2$

1. Consider a system containing the solvents octanol and water.
   a. Define/describe the equilibrium constant expression for the reaction of an analyte moving from water into octanol: $A_{water} \leftrightarrows A_{octanol}$. Use the symbol, *P*, for this partition coefficient.

   b. Theophylline and theobromine are structural isomers and are important methylxanthines in the food and beverage industry. The *log P* (P is the partition coefficient you defined above) values for theophylline (-0.17) and theobromine (-0.72) are given in the parentheses. If you had two separate liquid-liquid extraction systems with equal volumes of solvent (octanol and water) and start with 10.0 mM theophylline or theobromine in the water phase, what would be the concentrations in each phase after the extraction is performed?

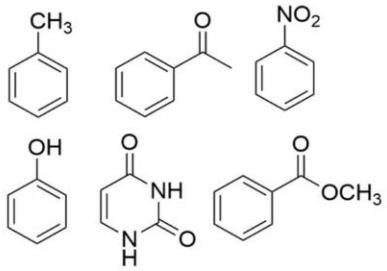
   c. For the separation in part b, what is the value of alpha, α, the separation factor?

   d. Define what you believe would be sufficient for "complete separation" for this octanol/water system if you start with 10 mM of both theophylline and theobromine in the water phase. Based on your definition, what would the value of α have to be to meet the requirements of your proposed definition.

   e. Using the value of α that you have defined and defended, describe how chromatographic techniques such as liquid chromatography are able to separate two very similar solutes such as theophylline and theobromine?

2. From a thermodynamic perspective, what drives any separation? The relationship for chemical potential might be useful to answer this question: $\mu_i = \mu_i^o + RT \ln C_i$.

3. The van Deemter equation, which relates the plate height, H, to various constants and mobile phase velocity, ν, is a common equation used in separations to compare new columns/column packings. Write this equation and define each symbol. Draw a blow-up sketch of a chromatographic column and illustrate the various contributions from each variable of the van Deemter equation to plate height.



4. Suggest and justify the order of elution for the following species on a C18 stationary phase with a 50:50 Methanol/Water mobile phase: in the Figure (Top, L to R) toluene, acetophenone, nitrobenzene. Bottom: L to R: phenol, uracil, methyl benzoate.

5.
   a. In gas chromatography (GC), what factors influence the choice of carrier gas?
   b. How does the choice of carrier gas affect separation efficiency and detection sensitivity?

6. A pharmaceutical formulation contains a mixture of acidic, basic, and neutral impurities. What separation strategies could be employed to analyze these impurities effectively?